

4.8 Centers of Gravity Using Multiple Integrals_Guide

Monday, January 12, 2026 10:42 PM



4.8 Centers
of Gravity...

4.8 Centers of Gravity Using Multiple Integrals

Motivation:

Now that we have the notion of double and triple integrals, we can easily get formulas for the center of mass of a lamina which does not have uniform density and center of mass of a three dimensional solid with varying density. We start by looking at the two-dimensional case.

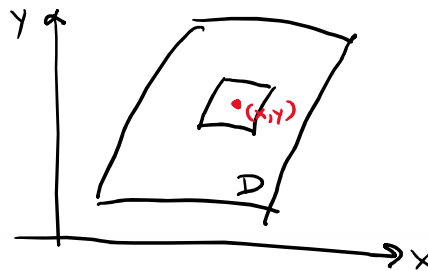
Suppose that a lamina occupies a region D of the xy -plane and its density (in mass per unit area since we view the lamina as two dimensional) at a point in D is given by $\rho(x, y)$ where ρ is a continuous function on D .

Note $\rho(x, y) = \lim \frac{\Delta m}{\Delta A}$

- Δm is the mass of a portion of the lamina containing (x, y)

- ΔA is the area of a portion of the lamina containing (x, y)

- the limit is taken as the dimensions of the rectangle approach 0.



So it makes sense to say $m = \iint_D \rho(x, y) \cdot dA$

Definition: The coordinates $(\bar{x}, \bar{y})$ of the center of mass of a lamina occupying the region D and having density function $\rho(x, y)$ are

$$\bar{x} = \frac{1}{m} \iint_D x \rho(x, y) dA \quad \text{and} \quad \bar{y} = \frac{1}{m} \iint_D y \rho(x, y) dA$$

Note: Three double integrals are required to find the center of mass of a lamina.

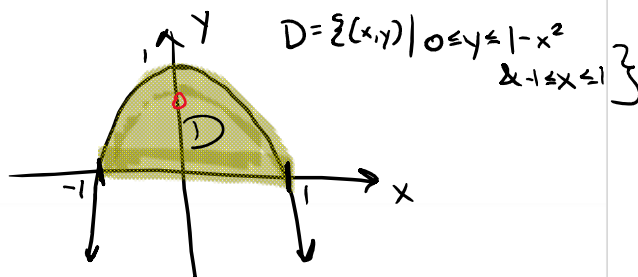
Note: Intuitively, one can view the double integrals for each coordinate of the center of mass as “weighted averages”.

Note: One will often hear $M_x = \iint_D y \rho(x, y) dA$ called the moment of the lamina about the x -axis, and will hear

$M_y = \iint_D x \rho(x, y) dA$ called the moment of the lamina about the y -axis.

Example 1: Suppose that a lamina occupies the region $D = \{(x, y) \mid y \leq 1 - x^2, y \geq 0\}$ and has density function $\rho(x, y) = y$. Find the center of mass of the lamina.

We sketch the region D .



Note

$$\text{mass} = \iint_D \rho(x, y) dA$$

$$= \int_{-1}^1 \int_0^{1-x^2} y dy dx = \int_{-1}^1 \left. \frac{1}{2} y^2 \right|_0^{1-x^2} dx = \int_{-1}^1 \frac{1}{2} (1-x^2)^2 dx = \int_{-1}^1 \frac{1}{2} (1 - 2x^2 + x^4) dx$$

$$= \frac{1}{2} \left(x - \frac{2}{3} x^3 + \frac{1}{5} x^5 \right) \Big|_{-1}^1 = \frac{1}{2} \left(1 - \frac{2}{3} + \frac{1}{5} \right) - \frac{1}{2} \left(-1 + \frac{2}{3} - \frac{1}{5} \right)$$

$$= \left(1 - \frac{2}{3} + \frac{1}{5} \right) = \frac{15 - 10 + 3}{15} = \frac{8}{15}$$

$$\bar{x} = \frac{1}{m} \iint_D x \cdot \rho(x, y) dA = \frac{15}{8} \int_{-1}^1 \int_0^{1-x^2} x y dy dx = \frac{15}{8} \int_{-1}^1 \left. \frac{1}{2} x y^2 \right|_0^{1-x^2} dx$$

$$= \frac{15}{8} \int_{-1}^1 \frac{1}{2} x (1-x^2)^2 dx = \frac{15}{16} \int_{-1}^1 (x - 2x^3 + x^5) dx$$

$$= \frac{15}{16} \left(\frac{1}{2} x^2 - \frac{1}{2} x^4 + \frac{1}{6} x^6 \right) \Big|_{-1}^1 = \frac{15}{16} \left(\frac{1}{2} - \frac{1}{2} + \frac{1}{6} \right) - \frac{15}{16} \left(\frac{1}{2} - \frac{1}{2} + \frac{1}{6} \right) = 0$$

$$\bar{y} = \frac{1}{m} \iint_D y \cdot \rho(x, y) dA = \frac{15}{8} \int_{-1}^1 \int_0^{1-x^2} y^2 dy dx = \frac{15}{8} \int_{-1}^1 \left. \frac{1}{3} y^3 \right|_0^{1-x^2} dx$$

$$= \dots$$

$$= \frac{4}{7}$$

So the center of mass is at $(0, \frac{4}{7})$.

We now explore how to find the center of mass of a three dimensional solid with varying density.

Definition: Suppose that the density function of a solid object that occupies the region E is the continuous function $\rho(x, y, z)$ (the unit for this function is mass per unit volume).

The center of mass of this object is $(\bar{x}, \bar{y}, \bar{z})$ where $\bar{x} = \frac{1}{m} \iiint_E x \rho(x, y, z) dV$, $\bar{y} = \frac{1}{m} \iiint_E y \rho(x, y, z) dV$, $\bar{z} = \frac{1}{m} \iiint_E z \rho(x, y, z) dV$, and $m = \iiint_E \rho(x, y, z) dV$.

Example 2: Find the center of mass of the solid that occupies the region bounded by the graphs of $x = y^2$, $x = z$, $z = 0$, and $x = 1$ which has density function $\rho(x, y, z) = 1$.

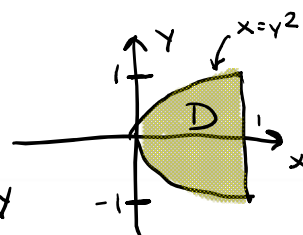
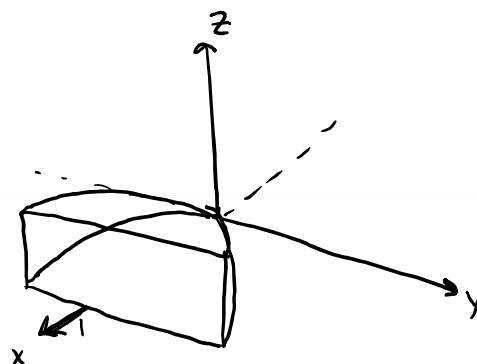
We first sketch a picture of E .

Note that for E , z is bounded by

$$0 \leq z \leq x$$

We then project into the xy -plane & get D .

$$D = \{(x, y) \mid y^2 \leq x \leq 1 \text{ \& } -1 \leq y \leq 1\}$$



$$\text{So mass} = \iiint_E \rho(x, y, z) dV = \int_{-1}^1 \int_{y^2}^1 \int_0^x 1 dz dx dy$$

$$= \dots = 4/5$$

$$\bar{x} = \frac{1}{m} \iiint_E x \rho(x, y, z) dV = \frac{5}{4} \int_{-1}^1 \int_{y^2}^1 \int_0^x x dz dx dy = \dots = \frac{5}{7}$$

$$\bar{y} = \frac{5}{4} \int_{-1}^1 \int_{y^2}^1 \int_0^x y dz dx dy = \dots = 0$$

$$\bar{z} = \frac{5}{4} \int_{-1}^1 \int_{y^2}^1 \int_0^x z dz dx dy = \dots = \frac{5}{14}$$

Center of mass $\left(\frac{5}{7}, 0, \frac{5}{14}\right)$